

Human vs. Shared Attribute Perceptions (#183068)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has 3 authors.

Pre-registered on: 07/15/2024 12:44 PM (PT)

1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

We are examining whether the relationship between algorithmic knowledge and preference for tasks to be completed by AI vs humans is moderated by the extent to which human attributes (vs. shared attributes) are typically required to complete the task.

3) Describe the key dependent variable(s) specifying how they will be measured.

In this study, we will be measuring perceptions of human vs. shared attributes required to complete 26 different tasks (1-Definitely have distinctively human attributes 7-Definitely have attributes shared between humans and AI). The 26 will be those used in the existing study (AsPredicted #126149).

4) How many and which conditions will participants be assigned to?

There will be no conditions. Attribute perceptions will be used as a predictor in a new analysis for the existing study (AsPredicted #126149)

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will calculate participants' average perceptions of human vs. shared attributes for each of the 26 tasks. This mean value will be included in the following analysis: We will run a mixed regression model, regressing AI preference for each individual task on participants' mean-centered AI literacy score, the mean-centered human vs. shared attribute score, and their interaction, with a random intercept for each participant to account for the repeated measures.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We will include one attention check. This question will be presented as an image and ask people to answer what 12-4 equals. Participants who do not write 8 will not be allowed to complete the study.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We will post the survey for 100 participants.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We have a study showing that there is a negative relationship between AI literacy and AI receptivity across 26 tasks (AsPredicted #126149). We plan to collect new data measuring perceptions about the 26 tasks as typically requiring human vs. shared attributes (which was not measured in the original study) to examine whether this moderates the relationship found in our existing data.